COS 30049 Computing Technology Innovation Project

Week4: Your AI Project‘s Journey

-- Machine Learning Project Lifecycle Introduction

Objectives:

1. Connect the dots between ML basics and your cybersecurity topic.
2. Basics of Machine Learning
3. Map out your project's roadmap (The ML Lifecycle).
4. Frame your idea as a clear, solvable ML problem.
5. Get started with the most critical phase: **Data Processing.**

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Why You Need to Think Like a Security Analyst?

To build a great AI model, you first need to understand the problem you're trying to solve. For your projects, that problem is rooted in cybersecurity.

Cybersecurity is about protecting data, systems, and users from threats.

Your Mission: Think Like a Security Analyst

- You'll be using real, high-impact datasets from cybersecurity domains.
- Your goal is to use ML to find attack patterns humans might miss
- Your projects simulate real industry challenges you'd face in industry.

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Why Domain Knowledge is Your Superpower

Why does it matter? Building a great AI model starts with understanding the problem you're solving. For you, that problem is in cybersecurity.

What are you looking for? Knowing the "signature" of a threat is what helps you find the right data and engineer the best features.

Analogy: You can't teach a model to find a needle in a haystack if you don't know what a needle looks like.

To build an effective AI model, you need to understand the problem domain. This contextual knowledge helps you:

- Find the right dataset.
- Engineer meaningful features.
- Correctly interpret your results.

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Project Option 1: Prompt Injection Guardrail (A2)

Think Like an AI Red Teamer

Your goal is to decide whether a message is trying to hijack an AI assistant, even when the attacker has disguised it.

Problem: Attackers craft messages that push an AI assistant into ignoring its own instructions.

Data: Real jailbreak and ordinary prompts collected from Reddit, Discord and prompt-sharing sites.
ML Task: Classification
block vs allow.
Impact: Protects LLM-powered applications from being manipulated.

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Project Option 1: Prompt Injection Guardrail (A2)

Think Like an AI Red Teamer
Your goal: Spot messages that try to take control of an AI assistant

To get started, investigate these ideas:

What is an injection trying to do? Override the instructions, swap the persona, or extract the hidden system prompt.

Why are ordinary prompts hard to tell apart? Legitimate requests also begin with "you are now" or "act as".

How do attackers hide it? Leetspeak, zero-width characters, base64, letter spacing.

Search: "prompt injection", "OWASP LLM Top 10", "guardrail evasion".

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Project Option 2: AI-Generated Content Detection (A2)

Think Like a Forensic Linguist

Your goal is to decide whether a piece of writing came from a person or from a language model.

Problem: Generated text is used for academic misconduct, fake reviews and social engineering.

Data: Human-written and model-generated essays from several sources and several models.
ML Task: Classification
human vs generated.
Impact: Supports academic integrity and flags synthetic content.

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Project Option 2: AI-Generated Content Detection (A2)

Think Like a Forensic Linguist
Your goal: Tell human writing apart from generated writing

To get started, investigate these ideas:

What is burstiness? People vary sentence length far more than a model does. How would you measure that?

What does type-token ratio measure? Which phrases do models over-use?

How do you score a document that is part written, part generated? Mean, maximum and fraction-above-threshold all behave differently.

Search: "burstiness AI text", "stylometry", "AI text detection".

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Project Option 3: Phishing URL Detection (A2)

Think Like a Threat Analyst

Your goal is to judge whether a web address is impersonating a real site, using the address alone.

Problem: Phishing sites imitate real ones to steal passwords and card details.

Data: Hundreds of thousands of benign, phishing, defacement and malware URLs.
ML Task: Classification
safe vs malicious.
Impact: Warns a user before they click.

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Project Option 3: Phishing URL Detection (A2)

Think Like a Threat Analyst
Your goal: Judge a web address without ever loading the page

To get started, investigate these ideas:

Anatomy of a URL: scheme, subdomain, registered domain, TLD, path, query. Which parts can an attacker freely choose?

Why is paypal.com.login-secure.xyz not PayPal? Find where the real domain ends and the deception begins.

What is typosquatting? What is a homograph attack? Which TLDs are cheap and heavily abused?

Search: "URL lexical features", "typosquatting", "homograph attack".

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Project Option 4: Log Anomaly Detection (A2)

Think Like a Site Reliability Engineer

Your goal is to find the sessions in a system log that do not behave like all the others.

Problem: Failures hide inside millions of routine log lines.

Data: Logs from a 33-node Hadoop cluster, labelled normal or anomalous for each block.
ML Task: Classification
normal vs anomalous session.
Impact: Catches failures before they become outages.

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Project Option 4: Log Anomaly Detection (A2)

Think Like a Site Reliability Engineer
Your goal: Find the sessions that behave unlike the rest

To get started, investigate these ideas:

What is a log template? Why turn "Received block blk_123 of size 456" into "Received block <*> of size <*>"?

What groups lines into one session? Here the block id does it. Same block id, same story.

A normal session follows a predictable sequence. The strongest clue is often an event that is missing, not one that is present.

Search: "log parsing", "log template extraction", "Drain algorithm".

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Your Base Datasets (A2)

One dataset per topic. Download it yourself and cite it.
Raw input and a label only: building the features is your work.

Prompt Injection Guardrail
huggingface.co/datasets/xTRam1/safe-guard-prompt-injection
AI-Generated Content Detection
kaggle.com/datasets/thedrcat/daigt-v2-train-dataset
Phishing URL Detection
kaggle.com/datasets/sid321axn/malicious-urls-dataset
Log Anomaly Detection
zenodo.org/records/8196385

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Why This Matter?



- ✓ Every topic is a real cybersecurity challenge.
- ✓ Every dataset is messy and real, just like in the industry.
- ✓ This is your chance to apply ML to make a real impact.

- ✓ **First step:** Understand the problem.
- ✓ **Next step:** Get your hands dirty with data.

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Basics of Machine Learning

Machine learning (ML) is a branch of artificial intelligence (AI) and computer science that focuses on the using data and algorithms to enable AI to imitate the way that humans learn, gradually improving its accuracy.

In machine learning, computers apply statistical learning techniques to automatically identify patterns in data. These techniques can be used to make highly accurate predictions.

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Basics of Machine Learning

Different types of Machine Learning:

- **Supervised Learning:** In supervised learning, the model is trained on labeled data, meaning each input comes with a corresponding output. The goal is for the model to learn the mapping from inputs to outputs so that it can predict the output for new, unseen data. Common tasks include classification and regression.

- **Unsupervised Learning:** In unsupervised learning, the model is trained on unlabeled data, where the algorithm tries to identify patterns or structure in the data without specific output labels. Common tasks include clustering and dimensionality reduction.

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Basics of Machine Learning

Different types of Machine Learning:

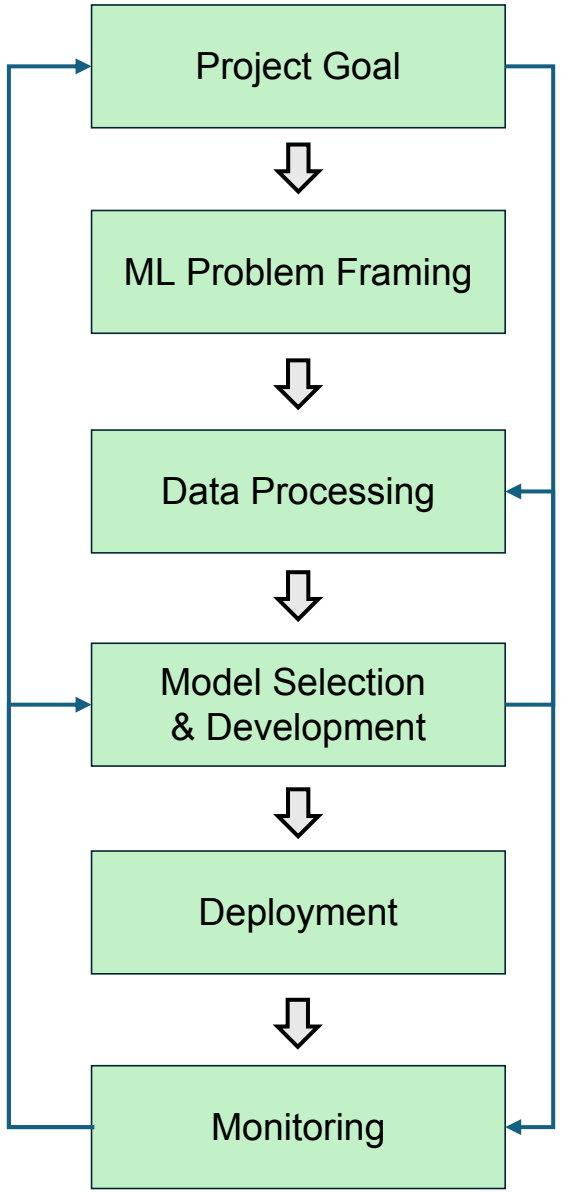
- **Semi-supervised learning:** The dataset contains structured and unstructured data, which guides the algorithm on its way to making independent conclusions. The combination of the two data types in one training dataset allows machine learning algorithms to learn to label unlabeled data.

- **Reinforcement learning:** The dataset uses a “rewards/punishments” system, offering feedback to the algorithm to learn from its own experiences by trial and error.

In this unit, we will only focus on Supervised Learning and Unsupervised Learning.

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Machine Learning Lifecycle



The ML lifecycle is a cyclic, iterative process that outlines best practices and guidelines for each phase of developing a machine learning workload. It provides clarity and structure, ensuring the success of a machine learning project.

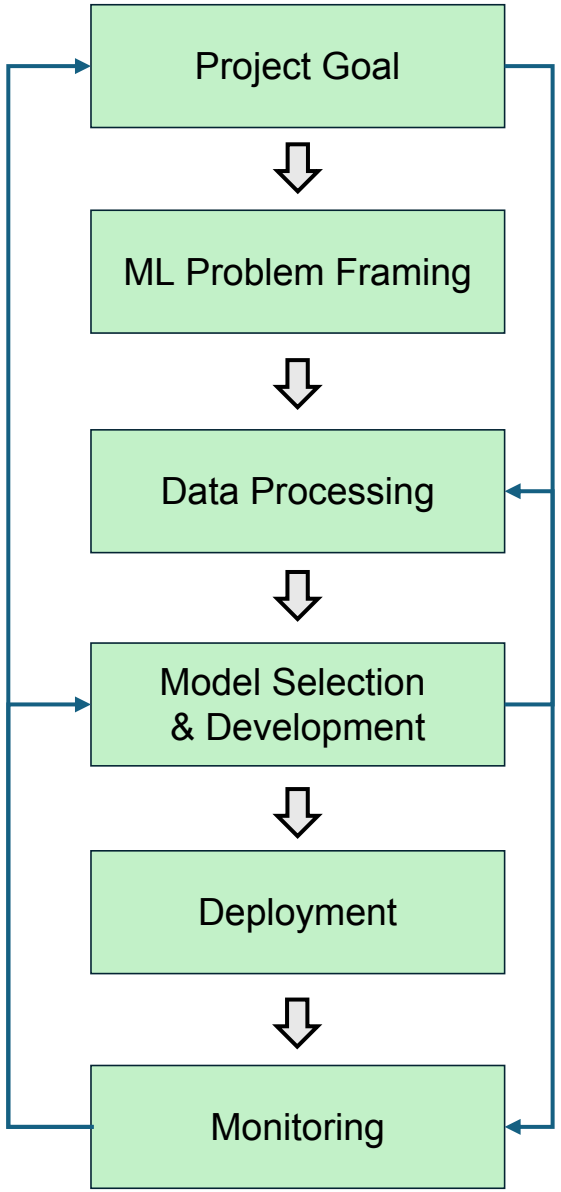
Project Goal Identification:

People Implementing ML should have a well-defined understanding of the problem they aim to solve and the value that solving this problem could bring. In this phrase, you will need to set your objective, and your metrics. For your project, it's essential to clearly identify the problem you have in mind and the outcomes you want to achieve.

ML Problem Framing: In this phase, the project's problem is translated into a machine learning context: identifying what data will be observed and what needs to be predicted (commonly referred to as the label or target variable).

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Machine Learning Lifecycle



Data processing

Training an accurate ML model requires data processing to convert data into a usable format.

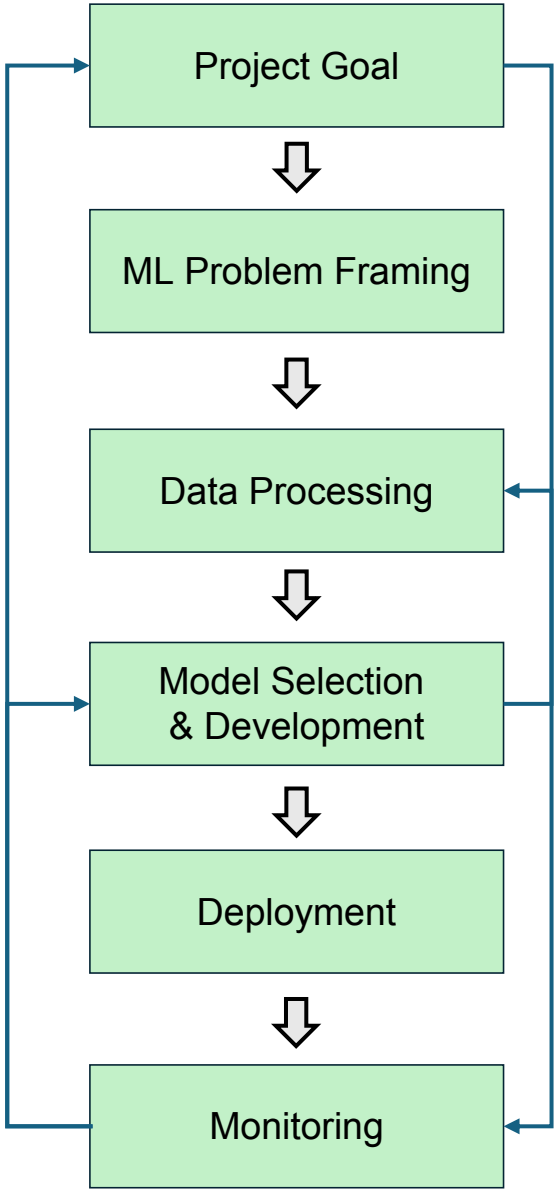
Data processing steps include collecting data, preparing data, and feature engineering that is the process of creating, transforming, extracting, and selecting variables from data.

Model selection and development

Model development consists of model building, training, tuning, and evaluation. The details of this part will be introduced in week 6-7.

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Machine Learning Lifecycle



Deployment

After a model is trained, tuned, evaluated and validated, you can deploy the model into production. You can use the model to make predictions and inferences.

Monitoring

Model monitoring system ensures your model is maintaining a desired level of performance through early detection and mitigation.

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Project Problem (A2)

Choose Your Topic

Select one of the four topics that interests you the most.

- Prompt Injection Guardrail
- AI-Generated Content Detection
- Phishing URL Detection
- Log Anomaly Detection

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Project Problem/Goal (A2)

Define the Problem/Goal

Identify a specific problem within your chosen topic that you want to address.

Formulate a clear and concise problem statement.

Misinformation Detection and Identification

Analyze online news articles, social media posts, and other digital content from the past decade to identify patterns and trends in misinformation. Develop a model to detect and classify misleading or false information, enabling organizations, media platforms, and the public to better identify and respond to misinformation in real time.

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Project Goal (A2)

The following work steps should be followed to establish your project goals.

- Understand project requirements.
- Form a project question.
- Identify critical, must-have features.
- Consider new project processes that might come out of this implementation.
- Consider how project value can be measured using metrics that the ML model can help to improve.

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ML Problem Framing (A2)

Problem framing is the process of analyzing a problem to isolate the individual elements that need to be addressed to solve it. Problem framing helps determine your project's technical feasibility and provides a clear set of goals and success criteria. When considering an ML solution, effective problem framing can determine whether your product ultimately succeeds.

Some view ML as a universal tool that can be applied to all problems. In reality, ML is a specialized tool suitable only for particular problems.

ML systems can be divided into two broad categories: predictive ML and generative AI. In this unit, we only learn and use predictive ML.

At a high level, ML problem framing consists of two distinct steps:

1. Determining whether ML is the right approach for solving a problem.
2. Framing the problem in ML terms.

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ML Problem Framing

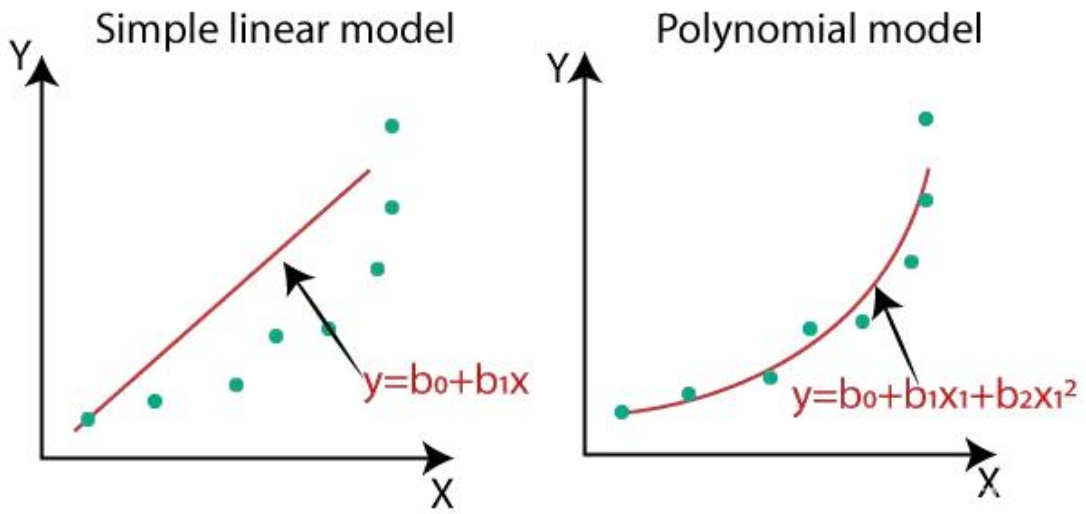
Machine learning can solve different types of problems:

Regression

Regression problems involve predicting the future values of continuous numerical variables.

For example, housing price prediction, stock price prediction, weather forecasting, etc.

Note: Assignment 2 requires classification and clustering. Regression is not required for any of the four topics.



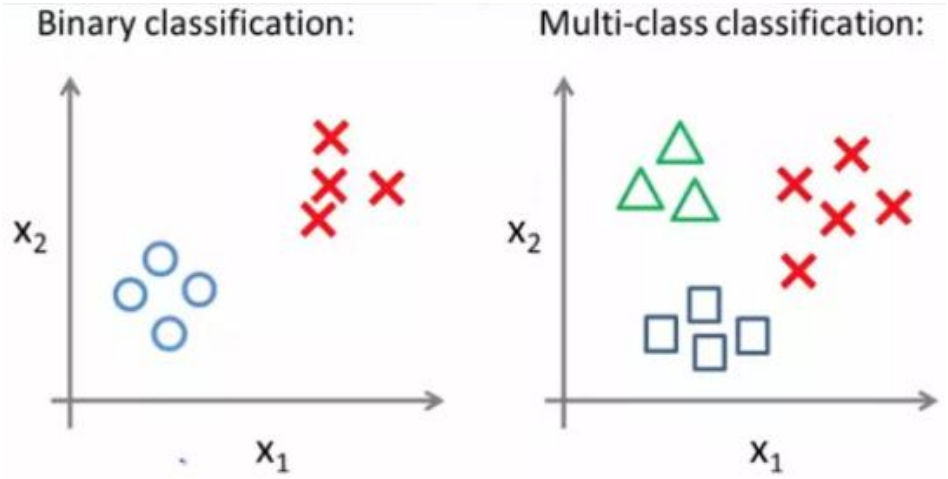
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ML Problem Framing (A2)

Classification

Classification problems involve categorizing data into different classes. For example, spam filtering (classifying emails as spam or non-spam), image classification (classifying images as cats, dogs, etc.), disease diagnosis (classifying patients into different disease types), etc.

(A2) Classification is the core detection task for every topic.

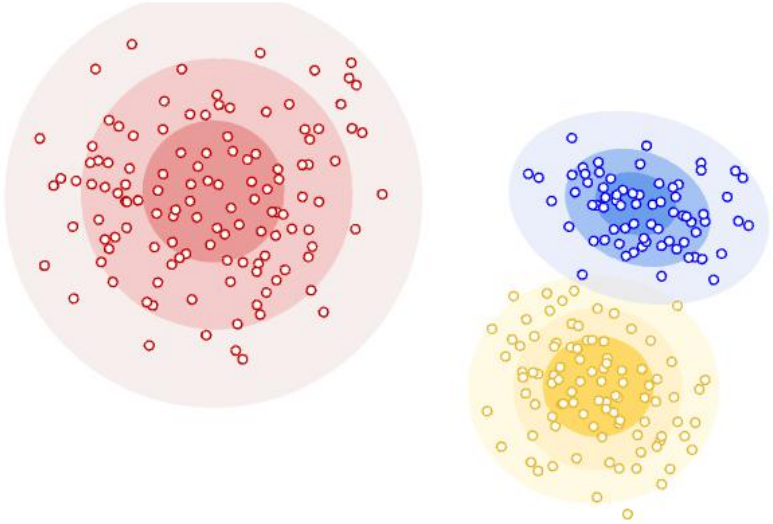


ML Problem Framing (A2)

Clustering

Clustering problems involve grouping data into different groups or clusters. For example, customer segmentation (grouping customers based on their behavior), image segmentation (dividing an image into different regions), etc.

(A2) Clustering is also required. Apply it within a single class and without using the labels.



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ML Problem Framing (A2)

Refine Project Objectives

First, analyze in detail what the project's goals are.

Do you need to predict a continuous value?

Do you need to classify data into different categories?

Do you need to discover hidden patterns or groupings in the data?

Examples:

1. Suppose you are working with a dataset that includes patient information from a healthcare system. The dataset is complex and includes both categorical and numeric features. You want to find patterns and similarities in the dataset. How might you approach this task?

2. A company wants to group its customers into segments based on their purchasing behavior (e.g., spending patterns and frequency of purchases). The goal is to find natural groupings within the customer base that were not previously defined.

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ML Problem Framing (A2)

Understand the Nature of the Data

Briefly review and analyze the data for the chosen topic.

Is the data continuous or discrete?

Does the data have labels (known outputs)?

How large is the dataset?

What are the number and types of features in the data?

Choose the type of ML problem

Classification?

Regression?

Clustering?

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Data Processing (A2)

Data processing is a broad concept that involves performing various operations on data to extract useful information. These operations include, but are not limited to:

- **Data Collection**: Gathering raw data from various sources.
- **Data Cleaning**: Removing or correcting errors, missing values, or inconsistent data.
- **Data Transformation**: Converting data from one format or structure to another.
- **Data Reduction**: Reducing the volume of data through aggregation or filtering while retaining important information.
- **Data Analysis**: Applying statistical methods and algorithms to analyze data and derive conclusions or identify patterns.

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Data Processing (A2)

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- **Data Reduction**: Reducing the volume of data through aggregation or filtering while retaining important information.
- **Data Analysis**: Applying statistical methods and algorithms to analyze data and derive conclusions or identify patterns.
- **(A2) A base dataset is provided for your topic. Finding additional data is optional and counts towards a higher grade.**

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Data Processing (A2)

Data Collection

Your data should have the following characteristics:

- ***Available.*** Make sure all inputs are available at prediction time in the correct format. If it will be difficult to obtain certain feature values at prediction time, omit those features from your datasets.
- ***Correct.*** In large datasets, it's inevitable that some labels will have incorrect values, but if more than a small percentage of labels are incorrect, the model will produce poor predictions.
- ***Representative.*** The datasets should be as representative of the real world as possible. In other words, the datasets should accurately reflect the events, user behaviors, and/or the phenomena of the real world being modeled. Training on unrepresentative datasets can cause poor performance when the model is asked to make real-world predictions.

If you can't get the data you need in the required format, your model will make poor predictions.

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Data Processing (A2)

Data Collection

Data Collection Methods

- Surveys: Use questionnaires to collect data online or offline.
- Experiments: Design and conduct experiments to collect data by controlling variables.
- Interviews: Conduct in-depth interviews via face-to-face, phone, or video.
- **Web Scraping**: Use automated scripts to scrape data from websites.
- **Sensor Data/Log Data**: Collect data using various sensors like temperature sensors, motion sensors, GPS, etc.; Collect data from server, application, or website logs.
- **Public Datasets**

  Use publicly available datasets from governments, research institutions, or companies.

  Sources: Kaggle, UCI Machine Learning Repository, government data portals.

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Data Processing (A2)

Data Collection

Data Sources

- Internet and Social Media: Collect data via APIs or web scraping from internet and social media platforms.

  Sources: Twitter API, Facebook Graph API, Instagram API.
- Government and Public Institutions: Access public data released by governments and public institutions.

  Sources: data.gov, EU Open Data Portal, World Bank Data.
- Research and Academic Institutions: Use datasets published by research and academic institutions.

  Sources: UCI Machine Learning Repository, Kaggle, IEEE DataPort.
- Companies and Enterprises: Internal business data or data from partners.

  Sources: company databases, CRM systems, ERP systems.
- Market Research Firms: Purchase or subscribe to data and reports from market research firms.

  Sources: Nielsen, Gartner, Forrsteer.

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Data Processing (A2)

Data Cleaning

1. Handling Missing Values

- Deleting Missing Values: If the proportion of missing values is small, consider deleting records that contain missing values.
- Imputing Missing Values: Use statistical methods such as mean, median, mode, or other techniques to fill in missing values.
- Interpolation: Use interpolation methods, such as linear interpolation, to fill in missing values.

2. Handling Duplicate Values

- Deleting Duplicate Values: Check for and delete duplicate records in the dataset.

3. Handling Outliers

- Box Plot Method: Use box plots to identify and handle outliers.
- Z-Score Method: Identify and handle outliers by calculating the Z-score.

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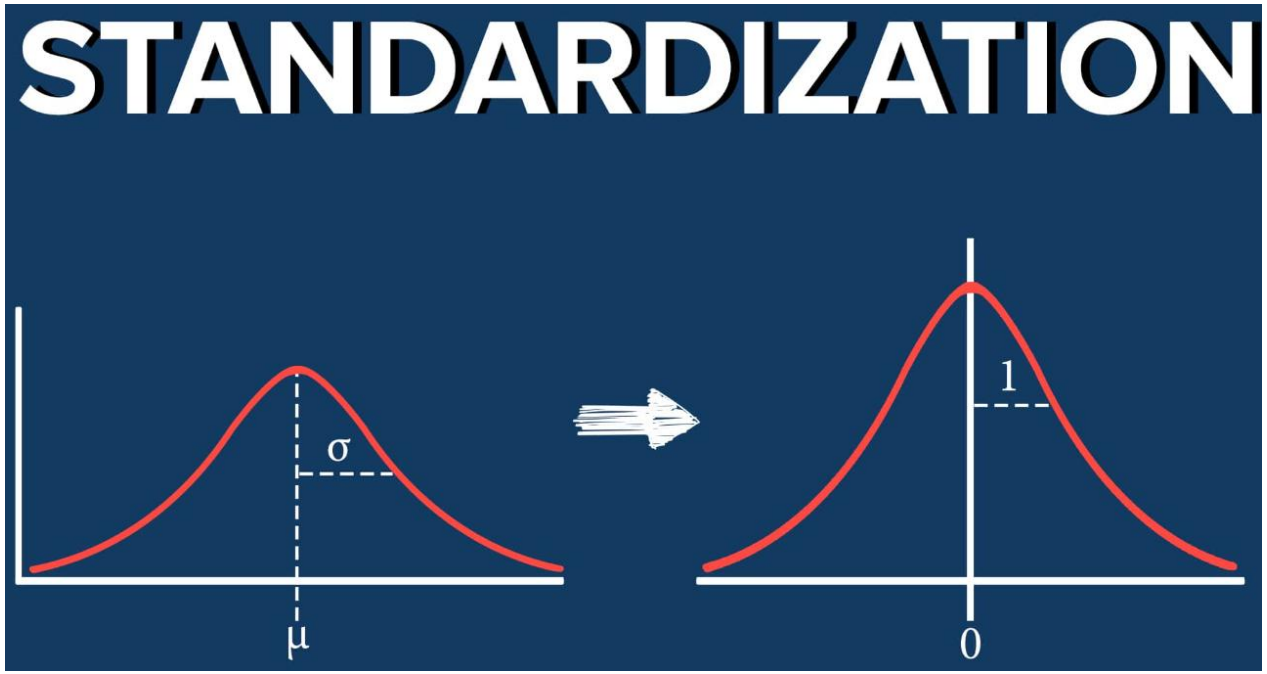
Data Processing (A2)

Data Transformation

1. Standardization: Converts features to have a mean of 0 and a standard deviation of 1.

$$x' = \frac{x - \mu}{\sigma}$$

Use Case: Suitable for features with different scales, like in neural networks and linear regression.



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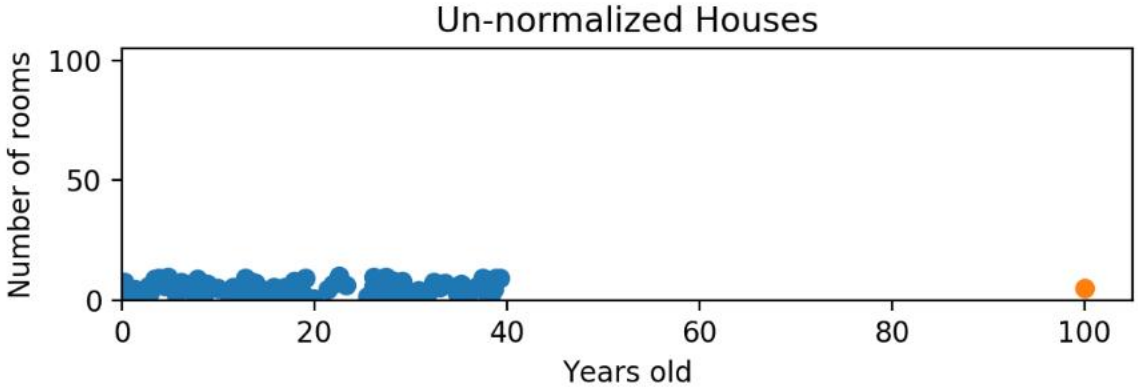
Data Processing (A2)

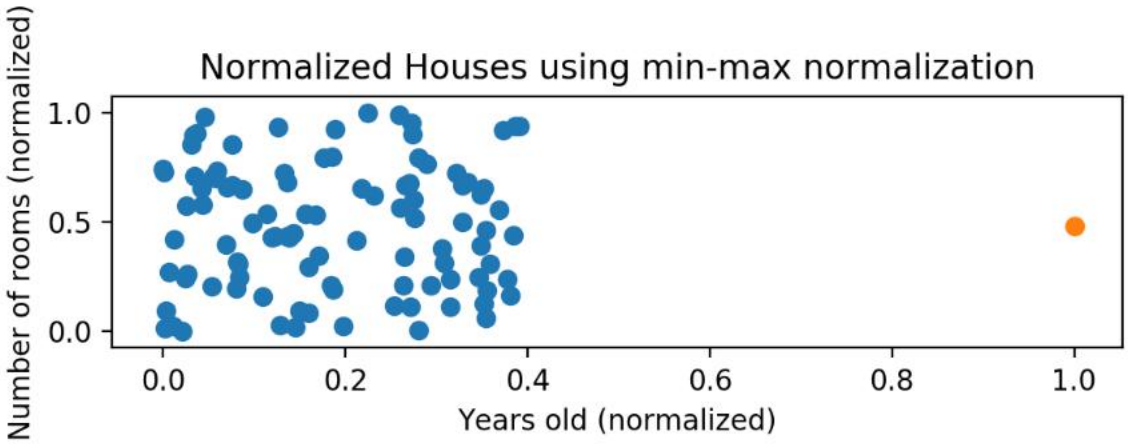
Data Transformation

2. Normalization: Scales features to a specific range, usually [0, 1].

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

Use Case: When features need to be on the same scale, such as in KNN and clustering algorithms.





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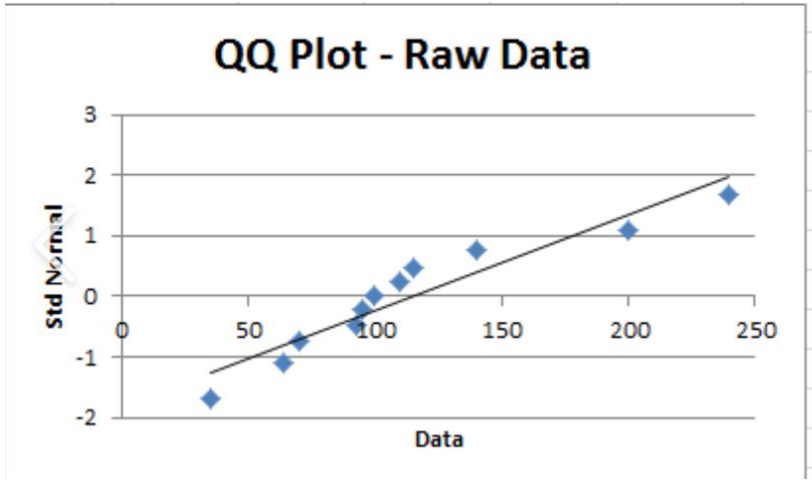
Data Processing (A2)

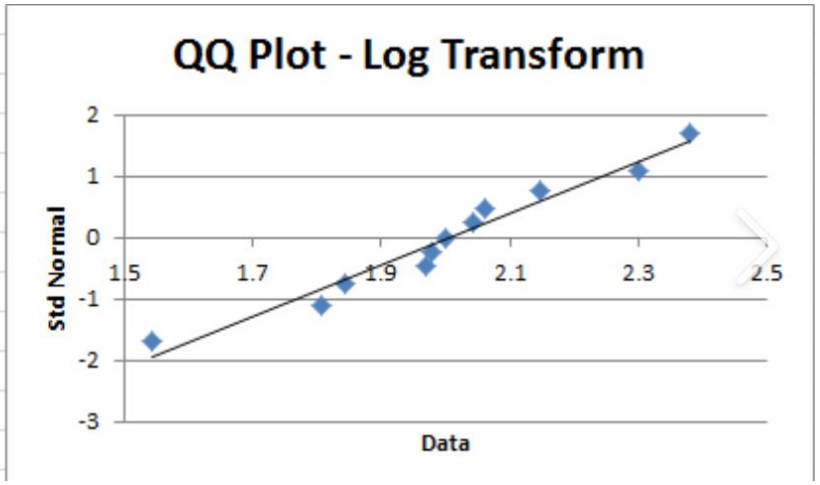
Data Transformation

3. Log Transformation: Converts skewed data into a more normal distribution.

$$x' = \log(x)$$

Use Case: Suitable for data with exponential growth or multiplicative effects.





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Data Processing (A2)

Data Transformation

4. Mean Deviation Transformation

Centers data by subtracting the mean from each data point.

$$x' = x - \mu$$

Use Case: To remove mean shift and center data distribution.

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Data Processing (A2)

Data Transformation

Transformation Method	Definition	Applicable Algorithms	Characteristics	Use Cases
Standardization	Adjusts data to have a mean of 0 and a standard deviation of 1, so features are on a similar scale.	K-Means, SVM, Linear Regression	Ensures features are comparable in scale; sensitive to outliers.	When features have different units or scales, and you need to compare them directly.
Normalization	Scales data to a fixed range, usually [0, 1] or [-1, 1], making all data fit within the same range.	Neural Networks (e.g., Deep Learning), KNN	Compresses data to the same range; ideal for algorithms sensitive to input scale.	When the input features have different ranges, and the model (like neural networks) requires data within a specific range.
Log Transformation	Applies a logarithm to data to reduce the impact of large values, making differences more manageable.	Linear Regression, Decision Trees	Reduces the effect of large values and skewness; not suitable for negative or zero values.	When data is highly skewed, such as income data, where large values dominate.
Mean Deviation Transformation	Shifts data around its mean, centering it at 0, which helps in identifying variations.	Time Series Analysis, Signal Processing	Centers data around the mean; useful for removing overall trends or shifts while retaining the original shape of data.	When you need to analyse data deviations from the mean, such as in signal processing or removing trends in time series data.

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Data Processing (A2)

Data Splitting

***Training Set**:* A dataset used to train the model. The model learns the relationships between features and labels in the training data, adjusting its parameters to correctly predict unknown data.

***Validation Set**:* A dataset used during training to evaluate the model's performance. It helps in tuning hyperparameters and selecting the best model, preventing overfitting.

Test Set: A dataset used for the final evaluation of the model's performance. It is completely independent, and the model's performance on the test set represents its expected effectiveness in real applications.

Why we need splitting?

Evaluating Model Performance: The test set is independent of the training process and is used to assess the model's performance on unseen data, thereby determining its generalization ability.

Improving Model Reliability: This three-way split allows for a more accurate assessment of the model's actual performance, ensuring that it performs well in real-world applications.

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Data Processing (A2)

Data Splitting

How to Split Train-Test

Steps:

1. **Divide the Dataset**:

   Comm on ratios are 70:30 or 80:20, meaning 70% or 80% of the data is used for training, and the remaining 30% or 20% is used for testing.

2. **Train the Model**:

   Use the training set to train the machine learning model.

3. **Evaluate the Model**:

   Use the test set to evaluate the performance of the model.

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Data Processing

Data Splitting – *Additional knowledge*

Cross-Validation

Purpose: To reduce bias in model evaluation and obtain a more stable and reliable performance assessment through multiple splits and training iterations.

Steps:

1. **Select a Cross-Validation Method:**

- k-Fold Cross-Validation: Divide the dataset into k subsets and iterate k times, using k-1 subsets for training and the remaining 1 subset for testing each time.
- Leave-One-Out Cross-Validation (LOOCV): Train the model on n-1 samples and test on the remaining 1 sample, repeating this for all samples.

2. **Compute Average Performance:**

- Calculate the average performance metrics (e.g., accuracy, mean squared error) across all iterations.

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Data Processing in Cybersecurity (A2)

Why it matters:

Cybersecurity data (logs, network traffic, binaries) is often **noisy, incomplete, and imbalanced.**

Preprocessing ensures data is clean, consistent, and ready for modeling.

Common challenges in security data:

Heterogeneous formats (PCAP, syslog, JSON, CSV)

Sensitive information (IP addresses, user IDs)

Class imbalance (few attack samples vs. many normal samples)

Example:

Raw network traffic may contain thousands of empty TCP packets → need to remove before analysis.

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Data Processing in Cybersecurity (A2)

Key Preprocessing Steps (Security-Specific)

1. Cleaning

Remove invalid packets/sessions (e.g., truncated packets in PCAP).

Synchronize timestamps across multiple log sources.

2. Feature Extraction

From network traffic: packet size stats, session duration, protocol types.

From logs: IP, domain names, file hashes, URLs.

From binaries: byte n-grams, API call sequences.

3. Anonymization (Privacy)

Mask IPs (e.g., replace 192.168.1.10 with hashed ID).

Replace user IDs/emails with random tokens.Slide 2 – Key Preprocessing Steps (Security-Specific)

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Data Processing in Cybersecurity (A2)

Scaling, Balancing, and Storage

4. Normalization & Standardization

Scale features like byte size or port numbers to prevent large values from dominating.

5. Handling Imbalanced Data

Oversample minority attack samples (e.g., SMOTE).

Undersample majority normal samples.

Apply cost-sensitive learning to increase weight of rare attacks.

6. Efficient Storage

Save processed features in Parquet/HDF5 for fast reading.

Partition logs by date/source system.

Example:

If attack logs are only 1% of dataset, use SMOTE to synthetically create more attack samples for training.

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Have a Break and Move to Lecture 02

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